Introduction to design evaluation and optimisation in PKPD

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CESU Advanced course in population approaches in pharmacokinetics/pharmacodynamics

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Overview

Population PKPD

- Population pharmacokinetic (PK) and pharmacodynamics (PD) studies increasingly performed for analysis of preclinical or clinical data
  - characterize PK/PD properties of a drug → supports the science underpinning the claimed therapeutic utility
  - Goal: sensitive study that maximizes the signal from any given change in the underlying system

- Several methods/software for maximum likelihood estimation of population parameters using nonlinear mixed effects models
  - NONMEM
  - Monolix
  - Pumas
  - R nlme, R saemix, etc.

- Problem beforehand: choice of 'population' design
  - number of individuals?
  - number of sampling times?
  - sampling times?
  - other design variables (doses, etc...)

Choice of design for Population PKPD analyses

- Increasingly important task for pharmacometricians
  - influence the precision of parameters estimation
  - poor design can lead to unreliable studies especially for complex models
  - all the more important in special population (paediatric studies ...)
    - severe limitations on the number of samples + ethical and physiological reasons
- Design considerations for population PK(PD) analyses in FDA and EMA guidelines

FDA Guideline (1999) [1]

Because it will determine the study design, the objective of a population PK study should be defined clearly from the outset. When designing a population PK study, practical design limitations, such as sampling times, number of samples per subject, and number of subjects, should be considered. Obtaining preliminary information on variability from pilot studies makes it possible through simulation (see section C, below) to anticipate certain fatal study designs, and to recognize informative ones. Optimizing the sampling design becomes particularly important when severe limitations exist on the number of subjects and/or samples per subject (e.g., in pediatric patients or the elderly) (15). Use of informative designs for population PK studies is encouraged (15-20). Such designs should include enough patients in important subgroups to ensure accurate and precise parameter estimation and the detection of any subgroup differences.

FDA Guideline (2022) [2]

Simulations and optimal design methods can maximize the utility of population PK data collection and analyses. For example, the trial sample size and PK sampling schedule can be optimized so that the PK parameters and the estimates for major **covariate** (bolded terms are defined in section VIII upon first use) effects of interest can be estimated with a defined degree of precision (see section IV.B for a discussion on various sampling schedules). Simulations with population PK models can help determine the number of patients in a subpopulation that achieves sufficient power to detect a significant covariate given a defined covariate effect size (e.g., the number of patients receiving a concomitant medication that should be included in an analysis to detect a significant drug-drug interaction (DDI); see section III.B.2).

EMA Guideline for PK in the paediatric population (2006) [3]

- "Population pharmacokinetic analysis, using non-linear mixed effects models, is an appropriate methodology for obtaining pharmacokinetic information in paediatric trials both from a practical and ethical point of view. The population approach may replace conventionally designed pharmacokinetic studies with rich sampling. **Simulations or theoretical optimal design approaches**, based on prior knowledge should be considered as tools for the selection of sampling times and number of subjects..."

1. FDA (1999) *Guidance for Industry Population Pharmacokinetics*
2. FDA (2022) *Guidance for Industry Population Pharmacokinetics*
3. EMA (2006) *Guideline on the role of pharmacokinetics in the development of medicinal products in the paediatric population*

Ju, G. *et al.* Optimized Sampling Strategies for Isoniazid in East Asian Pediatric Populations Using Population Pharmacokinetics-Informed Approaches. *Drug Design, Development and Therapy,* 3555–3576 (2025)

Drug Design, Development and Therapy

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ORIGINAL RESEARCH

Optimized Sampling Strategies for Isoniazid in East Asian Pediatric Populations Using Population Pharmacokinetics-Informed Approaches

Gehang Ju[1–3], Xin Liu[4], Yeheng Peng[4], Wenyu Yang[4], Nuo Xu[4], Qingfeng He[4], Chenchen Zhang[5], Lulu Chen[3,6], Nan Yang[3,6,7], Gufen Zhang[3,6,7], Chao Li[3,6,7], Pan Su[8], Xiao Zhu[4], Dongsheng Ouyang[1–3,6]

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Design in PK/PD

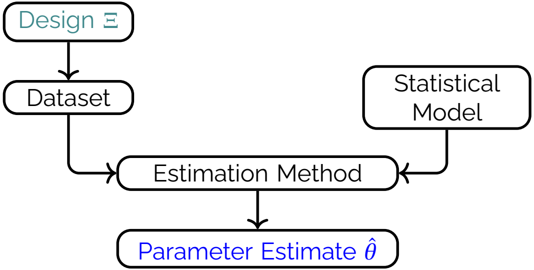
- **Individual designs**
  - standard regression $\neq Mixed-effectsmodel$
  - define number of samples (n) and sampling times (t_i)
- **Population designs**
  - nonlinear mixed-effects models
  - define number of patients (N) and number of elementary designs
    - for each elementary design: number of samples and sampling times, dosing regimen...
- **Bayesian designs**
  - Bayesian individual estimation
  - define number of samples (n) and sampling times (t_i)

Statistical estimation (1)

Statistics:

1. Inference
2. **Planning**

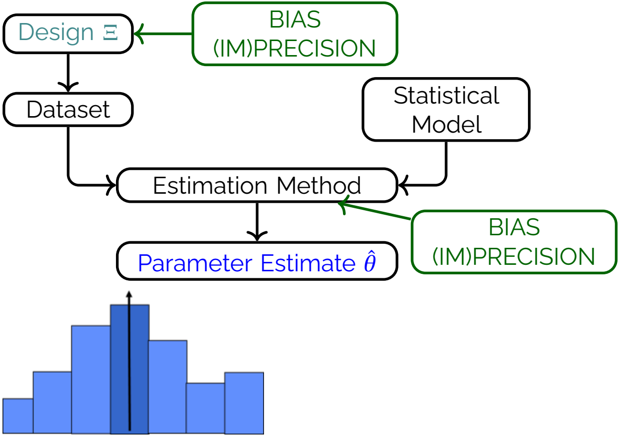
1. Inference
   - **estimation**
   - hypothesis testing
   - prediction
2. Planning = find 'optimal' design given
   - objective (e.g.: estimation)
   - statistical method (e.g.: maximum likelihood)
   - experimental constraints
   - some prior knowledge on expected results (e.g.: models and parameters)



Statistical estimation (2)

Same design

- Data set 1 → Estimate θ_1
- Data set 2 → Estimate θ_2
- ...
- Data set K → Estimate θ_K



Standard deviation of estimates

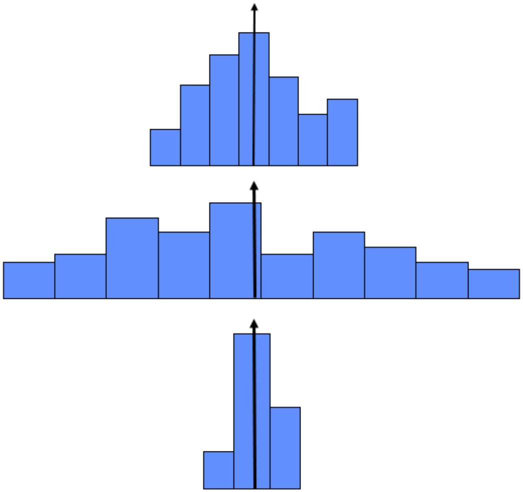
= standard error (SE)

= (im)precision

Statistical estimation (3)

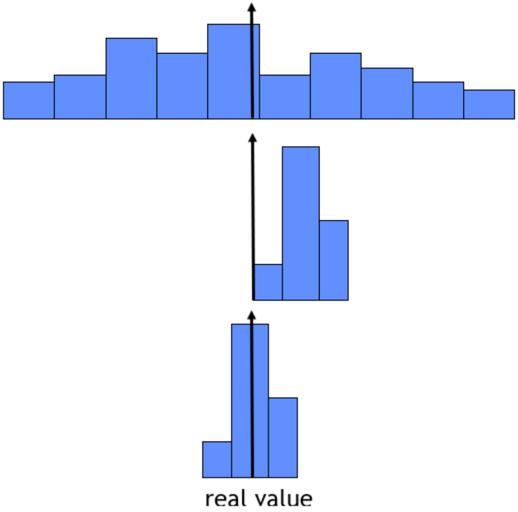
Find design with smallest SE:

- Design Ξ_1
- Design Ξ_2
- Design Ξ_3



Statistical estimation (4)

- Imprecise/Unbiased
- Precise/Biased
- Precise/Unbiased



Overview

Mixed effect model for one continuous response (1)

- N individuals $i = 1, \dots, N$
  - n_i observations $j = 1, \dots, n_i$
  - total number of observations: $n_{tot} = \sum_i n_i$

- j^{th} observation y_{ij} for the i^{th} subject:

$$y_{ij} = f(t_{ij},\ \xi_i,\ \theta_i) + g(t_{ij}, \xi_i, \theta_i, \sigma)\varepsilon_{ij}$$

  Vector notation: $y_i = f(\xi_i, \theta_i) + g(\xi_i, \theta_i, \sigma)\varepsilon_i$
  - y_{ij}: observation j in patient i
  - f, structural non-linear model
  - $\xi_i = \{t_i, d_i, \dots\}$: elementary design: vector of individual sampling times t_{ij}, dose d_i, ...
  - Individual parameters:

$$\theta_i = u\left(\mu, \beta, z_i, \eta_i\right)$$

    - μ, typical values
    - β, covariate effects
    - z_i, covariate vector
    - $\eta_i \sim \mathcal{N}(0, \Omega)$ individual random effects
  - Residual error
    - g
    - σ
    - random variable $\varepsilon_i \sim \mathcal{N}\left(0, I_{n_i}\right)$
  - $\Psi = \{\mu, \beta, \lambda, \Omega, \sigma\}$, P-vector of population parameters

Mixed effect model for one continuous response (2)

$$y_{ij} = f(t_{ij},\, \xi_i,\, \theta_i) + g(t_{ij},\xi_i,\theta_i,\sigma)\varepsilon_{ij}$$

- Error model: Residual unexplained variability (RUV)
  - $\varepsilon_i = \left(\varepsilon_{i1},\ldots,\varepsilon_{i,n_i}\right)$: zero mean random error, $\varepsilon_i \sim \mathcal{N}(0,\mathrm{I})$
  - Additive error model (homosedastic):

    $$g(t_{ij},\xi_i,\theta_i,\sigma) = \sigma_{inter} \quad \text{and} \quad \text{residual variance } = \sigma^2_{inter}$$

  - Proportional error model (heteroscedastic):

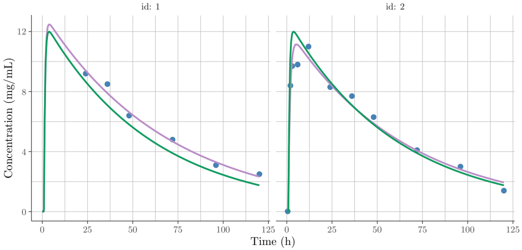
    $$g(t_{ij},\xi_i,\theta_i,\sigma) = \sigma_{slope} \times f(t_{ij},\, \xi_i,\, \theta_i) \quad \text{and} \quad \text{residual variance } = \left(\sigma_{slope} \times f(t_{ij},\, \xi_i,\, \theta_i)\right)^2$$

  - Combined error model (Combined1 in Monolix):

    $$g(t_{ij},\xi_i,\theta_i,\sigma) = \sigma_{inter} + \sigma_{slope} \times f(t_{ij},\, \xi_i,\, \theta_i) \quad \text{and} \quad \text{residual variance } = \left(\sigma_{inter} + \sigma_{slope} \times f(t_{ij},\, \xi_i,\, \theta_i)\right)^2$$

- Random-effects model
  - Between Subject Variability (BSV) / Inter-individual variability (IIV)
  - $\eta_i \sim \mathcal{N}(0,\Omega)$
  - Additive $\theta_i = \mu + \eta_i \longrightarrow$ Normal distribution: $\theta_i \sim \mathcal{N}(\mu,\Omega)$
    - If covariates: $\theta_i = \mu + \beta \times Z_i + \eta_i$
  - Exponential $\theta_i = \mu \exp(\eta_i) \longrightarrow$ Log-Normal distribution: $\log(\theta_i) \sim \mathcal{N}(\log(\mu),\Omega)$
    - If covariates: $\theta_i = \mu \exp(\beta \times Z_i + \eta_i)$

Basic mixed effect models-summary (3)



y_{ij}: observation j in patient i

$$y_{ij} = f(t_{ij}, \xi_i, \theta_i) + g(t_{ij}, \xi_i, \theta_i, \sigma)\varepsilon_{ij}$$

- Fixed and random effects → individual parameters
  - $\theta_i = \mu + \eta_i$
  - $\theta_i = \mu e^{\eta_i}$
- Fixed effects → mean parameters μ

Basic population design

- Elementary design ξ_i in individual i
  - number of samples n_i and sampling times: $t_{i1},\ldots,t_{in_i}$
  - may differ between individuals
  - may be sparse ($n_i < p$)
  - can also include dosing regimen

- Population design Ξ
  - set of elementary designs $\Xi = \{N, (\xi_1,\ldots,\xi_N)\}$
  - number of observations $n_{tot} = \sum n_i$
  - If one elementary design
    - N individuals
    - with same design ξ of n sampling times
    - $\Xi = \{N, \xi\}$
    - $n_{tot} = N \times n$
  - If Q elementary designs
    - Q groups of $N_q = w_q \times N$ individuals
    - Within each group same design ξ_q of n_q sampling times
    - $\Xi = \{N, (w_1,\ldots,w_Q), (\xi_1,\ldots,\xi_Q)\}$
    - $n_{tot} = \sum_q N_q n_q$

Design evaluation/optimisation for estimation

- Evaluate/compare designs given experimental constraints,
  - Predict standard error for each parameter
- Find optimal design
  - From given cost & experimental constraints
  - Smallest standard errors/ 'greatest' information in the data
- Two approaches
  - Simulation studies ('Clinical trial simulation')
  - Mathematical derivation of the Fisher Information Matrix

Evaluation of population designs by simulation

- **Strategy for each explored design:**
  1. Repeat S times:
     1. Simulate a dataset according to the chosen design Ξ
     2. Estimate the population parameter on this dataset, $\hat{\Psi}_s$
  2. Compute the expected uncertainty in the estimation as $sd(\hat{\Psi}_s)$
- Evaluation of population PK designs with respect to
  - number of patients (N), number of samples per patient (n)
  - sampling times
  - number of occasions per patient, number of samples per occasion

- Several published studies
  - Hashimoto, Y. & Sheiner, L. B. Designs for population pharmacodynamics: value of pharmacokinetic data and population analysis. *Journal of Pharmacokinetics and Biopharmaceutics* **19,** 333–353 (1991)
  - Jonsson, E. N. *et al.* Comparison of some practical sampling strategies for population pharmacokinetic studies. *Journal of pharmacokinetics and biopharmaceutics* **24,** 245–263 (1996)

- Main limitation
  - very time consuming
  - only limited number of designs evaluated

→ Approach without simulation: **population Fisher Information Matrix**

Fisher Information Matrix (1)

- Given ○ an elementary design ξ_i ○ a population parameter vector Ψ

$$M_F(\xi_i,\Psi) = \mathbb{E}\left(\frac{\partial \log L(y_i;\xi_i,\Psi)}{\partial \Psi}\frac{\partial \log L(y_i;\xi_i,\Psi)}{\partial \Psi^T}\right)$$

where $L(y_i;\xi_i,\Psi) = \int_{\mathbb{R}^d} p(y_i \mid \eta_i;\Psi\xi_i)p(\eta_i;\Psi)d\eta_i$

- Given ○ a population design $\Xi = \{N,(\xi_1,\ldots,\xi_N)\}$

$$M_F(\Xi,\Psi) = \sum_{i=1}^{N} M_F(\xi_i,\Psi)$$

- **Inequality of Rao-Cramer**: variance of any unbiased estimator bounded by the inverse of the FIM

$$cov(\hat{\Psi}) \geq M_F(\Xi,\Psi)^{-1}$$

  - standard errors (SE) on the parameters ≥ square root of the diagonal elements of M_F^{-1}
  - lower bounds usually close to values obtained for sufficiently rich designs

$\Longrightarrow$ **Expected SE resulting from a design can be computed from the FIM**

Fisher Information Matrix (2)

- If Q elementary designs

$$M_F(\Xi, \Psi) = N \sum_{q=1}^{Q} w_q M_F(\xi_q, \Psi) \quad \text{and} \quad cov(\hat{\Psi}) \geq \frac{1}{N} \left(\sum_{q=1}^{Q} w_q M_F(\xi_q, \Psi) \right)^{-1}$$

- If same elementary design ξ_i in all patients

$$M_F(\Xi, \Psi) = N \times M_F(\xi_i, \Psi) \quad \text{and} \quad cov(\hat{\Psi}) \geq \frac{1}{N} M_F(\xi_i, \Psi)^{-1}$$

$\Longrightarrow$ SE inversely proportional to $N^{1/2}$

$$SE(\Xi, \Psi)_p = \sqrt{\frac{1}{N} \left(\sum_{q=1}^{Q} w_q M_F(\xi_q, \Psi) \right)^{-1}_{pp}}$$

- Compute the uncertainty for another sample size
- Compute the required sample size to achieve a given confidence level

$$NSN(\Xi, \Psi, SE^\star)_p = N \left(\frac{SE(\Xi, \Psi))_p}{SE^\star} \right)^2$$

Fisher Information Matrix with discrete covariates

- Discrete covariate Z_i with C categories, $c_1, \ldots, c_C$
- p_c, proportion of subjects for which $Z_c = c_c$
- If the covariate distribution is independent of the design:

$$M_F(\Xi, \Psi) = N \sum_{q=1}^{Q} w_q \sum_{c=1}^{C} p_c M_F(\xi_q, \Psi, z_i = c_c)$$

Example: sex

$$M_F = p_{Male} M_F(Male) + (1 - p_{Male}) M_F(Female)$$

- FIM computation → power prediction for tests on covariate significance, or clinical relevance
- Optimise design to reach target power

FIM in Non-linear Mixed Effects Models

- No analytical expression for likelihood L neither for $M_F(\Xi, \Psi)$

- Approximation by the FIM of the Gaussian model obtained with a first order (FO) linearisation of the structural model [7] [8] [9] .
  - around a guess value of the population mean → block-diagonal FIM approximation

$$M_F(\xi, \Psi) = \begin{bmatrix} M_F(\xi, \mu, \beta) & 0 \\ 0 & M_F(\xi, \Omega, \sigma) \end{bmatrix}$$

  - around the mean 0 of the random effects → full FIM approximation.

- FO approximation
  - Implemented in several software (and giving same results) [10] [11]
    PFIM 7.0 [12] PopED 0.7.0. [13] $DESIGN tool [14] in NONMEM MlxDesignEval [15] in Monolix *OptimalDesign* [16] in Pumas
  - Better than FO for estimation
  - Limited if large variability and/or very nonlinear models
    → CTS or new methods (ex: Adaptive Gaussian Quadrature [17] [18] or Markov Chain Monte Carlo [19])

7. Mentre *et al.* (1997) *Biometrika*
8. Retout *et al.* (2002) *Statistics in medicine*
9. Bazzoli *et al.* (2009) *Statistics in medicine*
10. Nyberg *et al.* (2015) *British journal of clinical pharmacology*
11. Fayette *et al.* (2026) *Pharmaceutical Research*
12. R package version 7.0 , https://github.com/packagePFIM/PFIM
13. Foracchia *et al.* (2004) *Computer Methods and Programs in Biomedicine*
14. Bauer *et al.* (2021) *CPT: Pharmacometrics & Systems Pharmacology*
15. R package version 0.3.0
16. Tarek *et al.* (2021) *ACoP12*
17. Nguyen & Mentré (2014) *Computational Statistics & Data Analysis*
18. Ueckert & Mentré (2017) *Computational Statistics & Data Analysis*
19. Riviere *et al.* (2016) *Biostatistics*

Population design evaluation

- Local planification: parameters and model are given
- Objective: smallest SE = "smallest" M_F^{-1} or "largest" M_F

- Criteria for matrix comparison
  - **D-criterion:**, the most usual one: $\phi_D(\Xi, \Psi) = \left(det\left(M_F(\Xi, \Psi)\right)\right)^{1/P}$

- For a given model and a given Ψ and for each Ξ
  - predict M_F and D-criterion
  - predict SE or relative standard error RSE (%) for each population parameter
- Compare designs by ratio of D-criteria → gain or loss of efficiency

Population design optimisation

- Find design which maximises $det(M_F)^{1/P}$
- Design variables
  - continuous variables for the sampling times
    - within interval or within given set
  - discrete variables for the structure of the design
    - number of groups, of subjects per group, of samples per group

- Optimisation of exact or statistical designs
  - exact design
    - Fixed design structure (Q, N_q, n_q number of samples in ξ_q)
    - Optimisation of the sampling times in each group ξ_q, $q = 1, \dots, Q$
    - General algorithms: Simplex,... [20] ; [21]
  - statistical designs
    - optimisation of both the sampling times and the group structure
    - Fedorov-Wynn algorithm [7] [22] or Multiplicative algorithm [23]
    - sampling times in ξ_q among a given set (**discrete** times)
    - $w_q \approx N_q/N$ proportions of subjects in ξ_q
    - Q number of groups

20. Duffull *et al.* (2002) *Computer Methods and Programs in Biomedicine*
21. Retout & Mentré (2003) *Journal of Pharmacokinetics and Pharmacodynamics*
7. Mentre *et al.* (1997) *Biometrika*
22. Retout *et al.* (2007) *Statistics in Medicine*
23. Seurat *et al.* (2021) *Computer Methods and Programs in Biomedicine*

FIM extensions in NLMEM

- Several extensions of M_F
  - Multiple response models [9]
  - Inter-occasion variability, discrete covariates, power of tests [22] [24]
  - Covariance between random effects [25]
  - Bayesian FIM and shrinkage predictions [26]
  - Covariate distribution [27]
- Robust design accounts for model uncertainty [28]
- Adaptive 2-stage & robust designs [29]
- Covariate dsitrbution optimisation [30]

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9. Bazzoli *et al.* (2009) *Statistics in medicine*
22. Retout *et al.* (2007) *Statistics in Medicine*
24. Nguyen *et al.* (2012) *Statistics in Medicine*
25. Dumont *et al.* (2014) *Journal of biopharmaceutical statistics*
26. Combes *et al.* (2013) *Pharmaceutical research*
27. Fayette *et al.* (2024) *medRxiv*
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29. Fayette *et al.* (2023) *The AAPS Journal*
30. Fayette *et al.* (2026) *Computational Statistics and Data Analysis*

Bayesian FIM and shrinkage predictions

- **Goal:** estimate the individual random effects
  - Fixed effects are assumed to be known
  - e.g. Model-informed precision dosing (MIPD): propose sampling occasions to derive an individualized dose that results in effective and safe dosage regimens

- Bayesian FIM

$$M_{BF}(\xi_i,\eta_i) = \mathbb{E}\left(\frac{\partial \log p(\eta_i|y_i;\xi_i,\Psi)}{\partial \eta_i}\frac{\partial \log p(\eta_i|y_i;\xi_i,\Psi)}{\partial \eta_i^T}\right)$$

- Shrinkage Sh quantified by the ratio of the estimation variance predicted by M_{BF}^{-1} and the variance-covariance matrix of the random effects Ω

$$Sh = \text{diag}\left(M_{BF}(\xi_i,\eta_i)^{-1}\Omega^{-1}\right)$$

Optimal design robust to the model space

- Uncertainty on the form of the final model, e.g. a nonlinearity in the disposition phase is suspected
- More than one drug being administered at the same time, perhaps a drug–drug interaction study, and hence the design should perform well over both drugs

Goal: Find design that maximises:

$$\prod_{m=1}^{M} \left(det \left(M_F \left(m, \Xi, \Psi \right) \right) \right)^{1/P_m}$$

Optimal design robust to the parameter space

- Distribution of possible prior parameter values
  → Various optimisation criteria
  - e.g. maximising the expectation of the D-criterion over the prior of the parameters

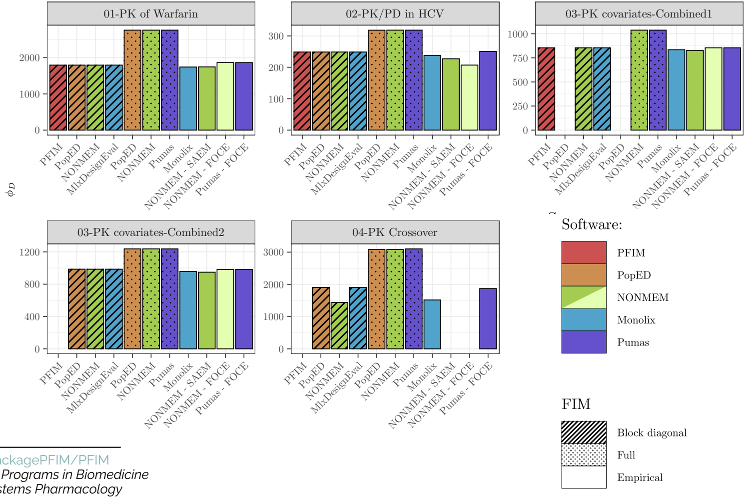
$$\mathbb{E}\left[\left(det\left(M_F\left(\Xi,\Psi\right)\right)\right)^{1/P}\right]$$

Overview

Software tools for population design approach

- PFIM 7 [12]
- PopEd 0.7.0 [13]
- NONMEM [14]
- *OptimalDesign* in Pumas [31]
- *MlxDesignEval* [15]

  → Comparison work [11]



12. R package version 7.0 , https://github.com/packagePFIM/PFIM
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14. Bauer *et al.* (2021) *CPT: Pharmacometrics & Systems Pharmacology*
31. Rackauckas *et al.* (2020)
15. R package version 0.3.0
11. Fayette *et al.* (2026) *Pharmaceutical Research*

Available features in the software tools available for population design evaluation

Feature	PFIM 7	PopED	NONMEM	Pumas	MlxDesignEval
Model					
Library of PK/PD models	Yes	Yes	No	No	Yes
User-defined models including ODE	Yes	Yes	Yes	Yes	Yes
Multiresponse models	Yes	Yes	Yes	Yes	Yes
BLQ Data	No	Yes	No	No	No
Fixed parameters	Yes	Yes	Yes	No	Yes
Variances					
Full covariance matrix for Ω	No	Yes	Yes	Yes	Yes
Full covariance matrix for Σ	No	Yes	Yes	Yes	No
IOV	No*	Yes	Yes	Yes	Yes
Residual error					
Additive or Proportional	Yes	Yes	Yes	Yes	Yes
Combined1 residual error	Yes	No	Yes	Yes	Yes
Combined2 residual error	No	Yes	Yes	Yes	Yes
Covariates					
Discrete covariates	No*	Yes	Yes	Yes	Yes
Power calculation	No	Yes	No	No	Yes
Design					
Multiple response designs	Yes	Yes	Yes	Yes	Yes
Designs differ across responses	Yes	Yes	Yes	No	Yes
FIM					
Block diagonal FIM	Yes	Yes	Yes	No	Yes
Full FIM	No	Yes	Yes	Yes	No
Prior FIM	No	Yes	No	No	Yes
Uncertainty in model parameters	No	Yes	Yes	No	Yes
Shrinkage (Bayesian FIM)	Yes	Yes	Yes	No	No
Plots					
Graphical visualisation of design	Yes	Yes	No	Yes	Yes
Graphical visualisation of sensitivity	Yes	No	No	No	No

No* indicates that the feature currently in development.

PFIM software

- Population Fisher Information Matrix
  - INSERM & University Paris Diderot registered software
  - Initially developed by Sylvie Retout and France Mentré in 2001
  - PFIM group (thepfimgroup@googlegroups.com)

- Using R, free, available at `www.pfim.biostat.fr`
- Last releases of PFIM
  - PFIM 4.0 in 2014 & PFIM Interface in 2015 [32]
  - R package PFIM 7 in July 2025
    `https://cran.r-project.org/web/packages/PFIM/index.html`

- Used in academia and industry to design new studies [33]

32. Dumont *et al.* (2018) *Computer methods and programs in biomedicine*
33. Mentré *et al.* (2013) *CPT: pharmacometrics & systems pharmacology*

Overview

Model and design

- PK model

$$f_{PK}\left(\theta^{PK},\xi^{PK}\right)=\frac{Dose}{V}\times\exp\left(\frac{-Cl}{V}\times\xi^{PK}\right)$$

  - θ^{PK}: Cl, V
  - Proportional error model

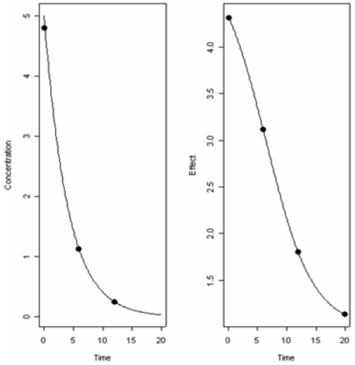
- PD model

$$f_{PD}\left(\theta^{PK},\theta^{PD},\xi^{PD}\right)=E_0+\frac{E_{max}\times f_{PK}\left(\theta^{PK},\xi^{PD}\right)}{C_{50}+f_{PK}\left(\theta^{PK},\xi^{PD}\right)}$$

  - θ^{PD}: E_0, E_{max}, C_{50}
  - Additive error model

(Example from Hooker, A. & Vicini, P. Simultaneous population optimal design for pharmacokinetic-pharmacodynamic experiments. *The AAPS journal* **7**, E759–E785 (2005))

- Population design
  - $N = 100$
  - $\xi^{PK} = \{0.167, 6, 12\}$
  - $\xi^{PD} = \{0.167, 6, 12, 20\}$



```
> getFisherMatrix(evaluationPopFim)$fisherMatrix
                      μ_V          μ_Cl        μ_E0        μ_C50        μ_Emax         ω²_V           ω²_Cl          ω²_E0           ω²_C50          ω²_Emax   σ_slope_RespPK   σ_inter_RespPD
μ_V            8049.06227   -4785.43296   171.45502     22.92870     -59.05664     0.000000       0.0000000       0.0000000        0.000000        0.000000          0.00000         0.000000
μ_Cl          -4785.43296  144709.18757    33.92710    731.80581     -97.07128     0.000000       0.0000000       0.0000000        0.000000        0.000000          0.00000         0.000000
μ_E0            171.45502      33.92710   845.98475    -89.46746      19.47652     0.000000       0.0000000       0.0000000        0.000000        0.000000          0.00000         0.000000
μ_C50            22.92870     731.80581   -89.46746    483.15905     -27.31367     0.000000       0.0000000       0.0000000        0.000000        0.000000          0.00000         0.000000
μ_Emax          -59.05664     -97.07128    19.47652    -27.31367      63.75449     0.000000       0.0000000       0.0000000        0.000000        0.000000          0.00000         0.000000
ω²_V              0.00000       0.00000     0.00000      0.00000       0.00000   518.299228      11.4501843       5.8793646        0.105145       11.160597        164.92387        30.163362
ω²_Cl             0.00000       0.00000     0.00000      0.00000       0.00000    11.450184     654.3984053       0.0143881        6.694247        1.884567         86.29946         9.413734
ω²_E0             0.00000       0.00000     0.00000      0.00000       0.00000     5.879365       0.0143881    3578.4509562       40.022133       30.346782        123.35506       994.958501
ω²_C50            0.00000       0.00000     0.00000      0.00000       0.00000     0.105145       6.6942467      40.0221329     1167.213345       59.682905        299.10382      1079.886002
ω²_Emax           0.00000       0.00000     0.00000      0.00000       0.00000    11.160597       1.8845665      30.3467819       59.682905     5202.732238         80.27361       245.133169
σ_slope_RespPK    0.00000       0.00000     0.00000      0.00000       0.00000   164.923871      86.2994593     123.3550608      299.103821       80.273606       2733.32007       523.746717
σ_inter_RespPD    0.00000       0.00000     0.00000      0.00000       0.00000    30.163362       9.4137341     994.9585005     1079.886002      245.133169        523.74672      9408.832026
```

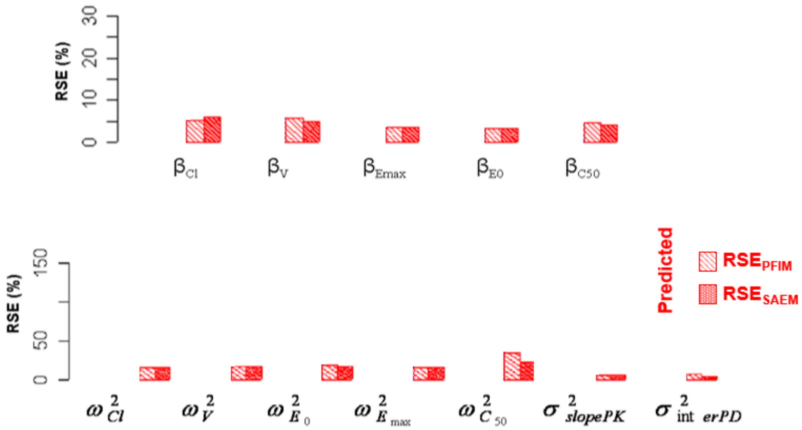
Evaluation of predicted SE

- Computation of the predicted standard errors with the expected Population Fisher Matrix
  - Relative Standard Errors: RSE_{PFIM}

- Comparison to the predicted SE obtained with an "exact method"
  - Simulation of one dataset with 10000 subjects
    - Computation of M_F by the SAEM algorithm (MONOLIX)
      → Asymptotic properties of M_F
    - Rescale of the SE for $N = 100$ subjects: RSE_{SAEM} [35]

35. Samson *et al.* (2007) *Statistics in medicine*

Predicted RSE (%): PFIM/SAEM



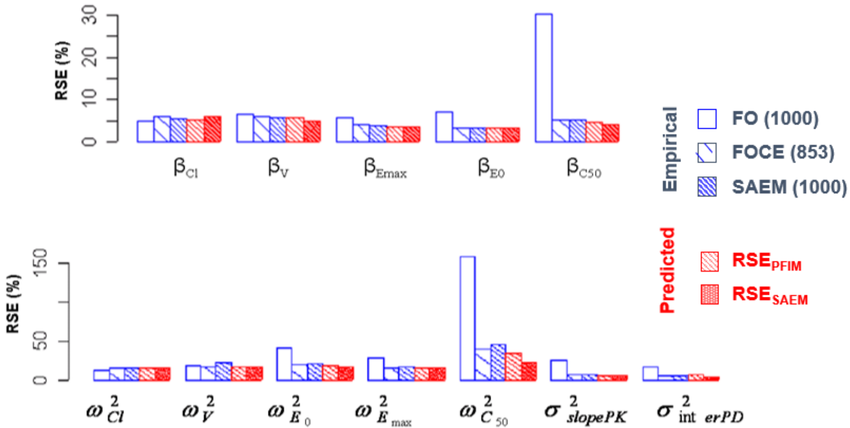
→ **Validate the accuracy of the linearisation to compute the FIM**

Comparison to empirical SE

- Simulation of 1000 datasets (R software)
- Estimation of the population parameters
  - NONMEM V (FO et FOCE)
  - MONOLIX v2.1: SAEM
- For each method of estimation:
  - Computation of the empirical RSE : standard deviation of the 1000 estimates of each parameter

- Convergence obtained for
  - FO: 1000 data sets
  - FOCE: 853 data sets (first-order conditional estimation)
  - SAEM: 1000 data sets

Empirical and Predicted RSE (%)



→ **Validate the accuracy of the FIM prediction**

Conclusion of these simulations

- Predicted RSE by PFIM
  - equivalent to the RSE predicted by SAEM
  - close to the empirical RSE of FOCE and SAEM from the 1000 replicates
  - Relevance of the first order approximation for design

- **Although the FIM computation is based on FO, the predicted RSE are close to those computed by FOCE and SAEM and not by FO**

Overview

Conclusions (1)

- Results of population PK/PD analyses increasingly used
  - in preclinical and clinical pharmacology
  - in drug labeling
  - in test of covariates
  - for clinical trial simulation

  $\Longrightarrow$ **Informative studies with small estimation error**
- Evaluation and comparison of population design without simulation using statistical approach based on the Fisher Information Matrix
- Results show that design may considerably affects precision of estimation

  Sparse sampling design → best information is needed
  Complex models → difficult to 'guess' good designs

Conclusions (2)

- Several software tools available: no excuses!
  - define good population designs (ethical/financial reasons)
  - anticipate 'fatal' population designs
  - Careful: lower bound (nonlinearity, small sample size)

- Local planification: relies on a priori information on models and parameters
  - sensitivity analysis with respect to population parameters and models
  - define compromise design, sampling windows...
  - use robust approach, adaptive design...

Conclusions (3)

- (Ongoing) work for stasticians/pharmacometricians
- FIM for joint (longitudinal and hazard) mixed models
- Better/faster optimisation algorithms (models/designs are more and more complex)
- Easier design optimisation tools
- Combined inference/design software $\Longrightarrow$ Increase use of model based design at all steps of drug development